

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and captured schematics for **optical control circuits**, including electro-mechanical components and signal chain processing, adhering to aerospace standards.
- Executed **Printed Circuit Board (PCB) layout** for complex optical readout circuits, incorporating high-speed layout principles for next-generation transceivers.
- Supported component selection processes, identifying and specifying parts suitable for **space-grade hardware** and demanding aerospace deployment environments.
- Developed and validated **power budgets** for critical subsystems, ensuring efficient energy distribution and management within stringent operational parameters.
- Engineered and tested prototype subsystems, progressing them through development cycles towards **aerospace-qualified products** and rigorous validation.
- Collaborated with mechanical and optical engineering teams to integrate electronic designs within mechanically and thermally constrained environments, meeting strict optical performance requirements.

SKILLS

Electronics Design – Analogue Circuit Design, Digital Circuit Design, Optical Control Circuits, Signal Chain Processing.

PCB Design & Simulation – Altium, KiCAD, SPICE Simulation, High-Speed Layout, RF Design Principles.

Aerospace Standards – AS/EN 9100, JEDEC Standards, Space-Grade Hardware, Aerospace Deployment.

Communications Protocols – CAN, SPI, UART, I2C, Free Space Optics (FSO).

Test & Validation – Laboratory Testing, Oscilloscopes, Multimeters, Noise Models, Link Budgets.

Power Management – Power Budgets, Subsystem Power Specification.

Software & Tools – Python, Java, SQL, Spring Boot, AWS, Docker, Linux, Git, Jenkins, Jira, Confluence, Power BI, Tableau.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Space-Grade Optical Transceiver Module

- Designed schematic capture for a **free space optics transceiver**, focusing on analogue readout and signal processing circuits like TIAs.
- Performed **SPICE simulations** to verify circuit performance and developed noise models and link budgets for the optical communication system.

- Collaborated with cross-functional teams to ensure mechanical and thermal compatibility for the design, targeting deployment in satellite communications systems.

Automated Test Platform for Control Systems

- Developed a Python-based automated test platform to validate the performance of **electronic control circuits**.
- Integrated with standard laboratory test equipment, including **oscilloscopes and multimeters**, to collect and analyze performance data.
- Documented test procedures and design validation reports, contributing to the overall system qualification process.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)